Outsourcing Dynamism

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Abstract

This paper investigates the increasing importance of domestic outsourcing in U.S. manufacturing. Under domestic outsourcing, the agency is the employer of record for temporary workers, though they perform their tasks at the client business' premises. On a yearly basis, one in two manufacturing plants hires at least some of its workers through a temporary help agency. Furthermore, domestic outsourcing is becoming increasingly more important: the average share of revenue spent on such arrangements has gone up by 85 percent since 2006. We develop a methodology to transform reported expenses on temporary and leased workers into plant-level outsourced employment counts, using administrative data on the U.S. manufacturing sector. We find that domestic outsourcing is an important margin of adjustment that plants use to modify their workforce in response to productivity shocks. Plant-level outsourced employment adjusts more quickly and is twice as responsive as payroll employment. These micro implications have significant aggregate consequences. Without taking reallocations in outsourced employment into account, the measured pace at which jobs reallocate across workplaces is underestimated. On average, we omit the equivalent of 15 percent of payroll employment reallocations in each year. However, outsourced employment churns at a much higher rate compared to its payroll counterpart. Therefore, the omission of outsourced reallocations can rationalize 37 percent of the secular decline in the aggregate job reallocation rate. Lastly, the extent of mismeasurement varies with the business cycle; falling in downturns and increasing in upturns implying that the speed of economic recovery is underestimated.

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1 Introduction

Traditionally, firms have met their labor needs by hiring workers directly. However, since at least the 1990s, U.S. firms have started hiring through third parties to "rent" workers—domestic outsourcing. This phenomenon of contracting other firms to meet labor demand is becoming increasingly more important: the U.S. manufacturing sector increased the number of outsourced jobs per payroll job by at least 40 percent between 2006 and 2017. Yet, previous data limitations have prevented this growing phenomenon from being incorporated into analyses of establishment-level and aggregate labor adjustment. This paper fills this gap by showing that (i) outsourced workers are an important margin of adjustment at the micro and aggregate level, and (ii) not accounting for the creation and destruction of jobs filled by outsourced workers biases the measurement of indicators at the center of our understanding of labor markets and the design of public policies targeting firms, e.g. job creation, job destruction, hires, separations, vacancies, and employment.

A large literature has documented a downward trend in the pace at which jobs and workers move across workplaces in the U.S. in recent decades (Bjelland, Fallick, Haltiwanger and McEntarfer, 2011; Decker, Haltiwanger, Jarmin and Miranda, 2016a; Molloy, Trezzi, Smith and Wozniak, 2016; Akcigit and Ates, 2021).[1] This decline in labor market fluidity has received considerable attention because less fluid labor markets result in lower productivity growth through the misallocation of resources: fewer jobs and workers are put to their most productive uses (Jovanovic and Moffitt, 1990; Haltiwanger, Foster and Krizan, 2001). Decker, Haltiwanger, Jarmin and Miranda (2020) find that reduced plant-level payroll employment responsiveness to revenue productivity underlies the aggregate decline in payroll job reallocations and has been a drag on aggregate productivity. Their results introduced allocative efficiency to the productivity slowdown debate and provided another channel behind the underlying causes of payroll job reallocations beyond changes in demographics.[2] However, the factors behind declining plant payroll employment responsiveness are still an open question and an important one given its implications for allocative efficiency.

This paper establishes that the increasing use of domestic outsourcing is one of the factors behind the decline in plant-level payroll employment responsiveness and accounts for a significant share of the aggregate decline in payroll job reallocations. The growing availability of labor market intermediaries has broadened the choice set for employers seeking to adjust employment. In particular, these intermediaries specialize in flexible labor sourcing, which is particularly attractive for plants responding to frequent, and possibly short-lived, productivity fluctuations.

[1] Although the level of the decline largely depends on the data source and the indicator used to measure it, there is a strong agreement about the downward trend (see Hyatt and Spletzer, 2013).

[2] The discussion about the documented slowdown in U.S. productivity growth focused on technological and measurement explanations (see, for example, Byrne, Fernald and Reinsdorf, 2016; Gordon, 2016; Syverson, 2017).

Domestic outsourcing happens when firms (or "clients") contract with other firms or individuals in the economy to provide goods and services previously performed in-house. Thus, outsourced staff effectively work for client firms but are legally employed by a staffing agency; therefore, these workers may change jobs, tasks, and workplaces but not the employer of record as in traditional employment relationships. While the U.S. data accounts for outsourced workers' labor market transitions across staffing agencies, it omits their reallocations across client establishments altogether —even in administrative sources. This omission gives rise to (i) a systematic undercount of aggregate job and worker reallocations (and the vacancies they fill), and (ii) a misrepresentation of job reallocation across sectors. These issues are the core of the *omitted reallocations* problem.

The biggest challenge to empirically investigate the implications of the omitted reallocations problem lies in the very nature of the problem at hand: *omitted* reallocations are unobservable. The client firm-outsourced employee relationship is not observable, and thus empirically documenting the relationship between outsourcing and business productivity has not been possible until now.[3] We overcome this challenge by combining multiple datasets administered by the U.S. Census Bureau and the U.S. Bureau of Labor Statistics, and developing a framework to compute plant-level outsourced employment from plant-level information on staffing services expenses.[4]

We begin our empirical analysis by characterizing the type of manufacturing plants using domestic outsourcing while documenting novel facts on the prevalence and growth of this phenomenon. First, we find that the share of revenue spent on outsourced workers (outsourced labor share) is decreasing in the plant's age and payroll employment size. These patterns are in line with the drop in the share of payroll jobs created by smaller and younger firms partly accounting for the long-term decline in labor market fluidity (Davis and Haltiwanger, 2014; Decker, Haltiwanger, Jarmin and Miranda, 2014; Decker, Haltiwanger, Jarmin and Miranda, 2016b). Second, between 2006 and 2017, the share of revenue spent on outsourced labor grew faster than that spent on payroll labor, respectively, by 85 percent and 10 percent for the average manufacturing plant. Third, the decision to use staffing arrangements as well as the intensity with which they use them vary systematically with revenue growth. The share of manufacturing plants using outsourced labor decreases when revenue growth is shrinking and increases when it is expanding. These three facts suggest that plants use domestic outsourcing strategically.

[3] See Bernhardt, Batt, Houseman and Appelbaum (2016) and Houseman and Bernhardt (2017) for a discussion on the data limitations to study outsourcing and an overview of the existing data.

[4] The Census and Annual Survey of Manufacturers collect plant-level information on expenses on different types of outsourced staff since 2006.[5] We focus on temporary and leased employees because this type of outsourced workers performs tasks on the client's worksite, typically in occupations at the core of the manufacturing business (Houseman, Dey and Polivka, 2010). From now on, we use temporary and leased employment and outsourced employment interchangeably for simplification.

Our findings indicate that outsourced employment is an important margin of adjustment.[6] We find that plant-level outsourced employment is twice as responsive as payroll employment to idiosyncratic shocks and adjusts more quickly. Plants respond to shocks by adjusting outsourced employment growth within the same year of the shock, while payroll employment reacts in the following year. This "immediate" response of outsourced employment reflects the flexible nature of this type of worker and is micro evidence for the use of temporary help employment as a leading recession indicator (see Peck and Theodore, 2007; Houseman and Heinrich, 2015; Luo, Mann and Holden, 2021). Following negative shocks, plants shed outsourced jobs first and retain payroll workers in whom they have invested and who have acquired firm-specific skills. If the shock proves persistent, then payroll jobs are also destroyed. On the other hand, following positive shocks, plants outsource new hires; waiting to see whether the shock is long-lasting enough to merit costly investments in recruiting, hiring, and training new payroll employees. These patterns are consistent with the hypothesis that payroll employment adjustment costs are an important mechanism behind the use of staffing services (Abraham and Taylor, 1996; Segal and Sullivan, 1997; Houseman, 2001; Autor, 2003; Houseman, Kalleberg and Erickcek, 2003).

Not only does domestic outsourcing affect micro-level firms' behavior, but it also has large implications for our understanding of aggregate outcomes. On average, every year, we omit the equivalent to 14 percent of payroll job reallocations. Moreover, the creation or destruction of outsourced jobs can account for as much as one-fifth of the corresponding payroll job flow in a given year; implying that the drop in total reallocations, both outsourced and those involving payroll jobs, is about 60 percent of the measured one. Omitting outsourced reallocation alters our understanding of labor markets' adjustment along the cycle, as well. For instance, in 2010, the manufacturing sector was creating jobs to be filled by outsourced workers at a higher pace than payroll jobs, and it also was not destroying the existing outsourced jobs as quickly as payroll jobs. This evidence indicates that the U.S. economy recovered at a faster pace from the Great Recession than previously documented.

ADDITIONAL CONTRIBUTIONS TO THE LITERATURE. Our findings contribute to two strands of literature: labor market dynamism and domestic outsourcing. Our findings indicate that the secular decline in job reallocation rates is substantially smaller than previously measured when outsourced jobs are taken into account. We also contribute to the literature on domestic outsourcing by highlighting the importance of domestic outsourcing *dynamics* and providing unique evidence from administrative plant-level data for the biggest client sector of staffing services, i.e. manufacturing.

[6] We build on Decker, Haltiwanger, Jarmin and Miranda's (2020) empirical design to study both payroll employment and outsourced employment responsiveness to revenue productivity changes. We define shocks as those authors do, that is revenue productivity growth deviations from own or detailed industry-year averages.

Domestic outsourcing is receiving increasingly more attention due to its consequences for wages, wage discrimination within the firm, and wage inequality (Houseman, Kalleberg and Erickcek, 2003; Autor and Houseman, 2010; Dube and Kaplan, 2010; Goldschmidt and Schmieder, 2017; Bloom, Guvenen, Smith, Song and von Wachter, 2018; Dorn, Schmieder and Spletzer, 2018; Bilal and Lhuillier, 2021; Bergeaud, Mazet-Sonilhac, Malgouyres and Signorelli, 2021; Weber Handwerker, 2022). Not surprisingly, then, this literature is mostly focused on the worker side. Two notable exceptions to this practice are Houseman (2001) and Anderson and McKenzie (2022). Our contribution to this literature is twofold. First, the descriptive evidence of this paper complements our knowledge about domestic outsourcing from the client business side. Our cross-sectional findings on plants' usage of staffing services are novel. Moreover, they are consistent with previous work (see Houseman, 2001).

Second, our results show that the dynamics in the use of domestic outsourcing also have significant consequences on our understanding of labor markets' functioning. The existing empirical evidence concentrates on the effects of transitioning from no outsourcing to outsourcing. This paper also adds the magnitude of the decline in gross job flows to the list of measurement issues that arise if we do not account for outsourced workers. Previous work has focused on employment and labor productivity patterns, or worker flows (Dey, Houseman and Polivka, 2012; Houseman and Heinrich, 2015; Dey, Houseman and Polivka, 2017; Atencio De Leon, 2023).

Accounting for domestic outsourcing complements our knowledge of the decline in labor market dynamism and of labor adjustment at the aggregate and micro level (Shimer, 2012; Hyatt and Spletzer, 2013; Davis and Haltiwanger, 2014; Decker, Haltiwanger, Jarmin and Miranda, 2016a; Molloy, Trezzi, Smith and Wozniak, 2016; Peters and Walsh, 2019; Decker, Haltiwanger, Jarmin and Miranda, 2020; Akcigit and Ates, 2021). Our results suggest that at least part of the decline reflects a transformation of the labor market towards the use of intermediaries in the employment process rather than a decline in underlying dynamism and that allowing a broader definition of employers' responsiveness to include domestic outsourcing enriches our understanding of micro-level adjustments to shocks.

OVERVIEW OF THIS PAPER. Section 2 documents the prevalence and growth of domestic outsourcing in the U.S. manufacturing sector. Section 3 presents the methodology to estimate plant-level outsourced employment from expenses and assesses the consequences of omitting outsourced workers on plant-level labor responsiveness of plants to changes in revenue productivity. Section 4 quantifies the omitted reallocations problem in manufacturing job reallocations. Section 5 concludes.

2 Prevalence of domestic outsourcing in U.S. manufacturing

This section describes domestic outsourcing in the U.S. manufacturing sector using administrative micro data containing expenses on temporary and leased staff for the period between 2006 and 2017 (see section 2.1). We describe the prevalence of domestic outsourcing by plant size and age, high/low-tech industry, three-digit industry, and revenue growth. We also characterize the types of plants that use outsourced staff more intensively, as measured by the share of revenue spent on this type of workers. Finally, we investigate the use of outsourced employment over time. The results in this section show that domestic outsourcing is increasing over time and exhibits substantial cross-sectional variation along plants' characteristics and revenue growth.

2.1 Data

Large data gaps have impeded accurately accounting for outsourced workers in economic analyses. We overcome this challenge by combining multiple administrative datasets from the U.S. Census Bureau. Specifically, we use the Annual Survey of Manufacturers (ASM), the Census of Manufacturers (CM), the Longitudinal Business Database (LBD), and the Revenue-Enhanced Longitudinal Business Database (RE-LBD).

The Annual Survey of Manufacturers contains a sample representative of the manufacturing sector that rotates every five years (in years ending in "4" and "9"). The Census of Manufacturers, on the other hand, contains the universe of manufacturing plants and is conducted in years ending in "2" and "7". For such years, We keep only the plants that are in the corresponding rotating sample of the ASM. The rotating sample feature of the ASM is essential for our empirical analysis since we rely on year-to-year changes.

The ASM-CM has establishment-level information on revenue and revenue productivity and since 2006, on expenses on outsourced services[7]. We focus on expenses on temporary and leased staff since, in manufacturing, this type of outsourced workers typically work side-by-side payroll employees and are in production occupations (Dey, Houseman and Polivka, 2017).[8] These characteristics of temporary and leased workers are important for our analysis because they suggest that the jobs filled by these types of outsourced workers are comparable to those filled by payroll employees. This implies that the omission of the creation and destruction of jobs filled by temporary and leased workers reflects a structural change in the employment process (the use of intermediaries)

[7]Revenue is defined as the sum of the total value of shipments and variations in inventory.

[8]Domestic outsourcing includes independent contractors, on-call workers, contract company workers, temporary workers, and leased staff. Temporary help agencies assign their workers to client plants, while professional employment organizations (the legal employer of leased employees) completely take over client plants' human resources tasks.

rather than in the task composition of the production process. From now on, we will use temporary and leased workers and outsourced workers interchangeably.

We link the ASM-CM sample with the LBD to retrieve information on plant location, plant age, firm age, and firm payroll employment. The LBD is a census of establishments and firms in the U.S. with paid employees. We use the plant location to link each plant with the average labor share (payroll over revenue) of temporary help firms and professional employment organizations in a given state. This information is in the RE-LBD and is an important component of the estimation of outsourced staff from expenses on temporary and leased staff.

We restrict the ASM-CM sample to establishments with no missing or imputed information on expenses on temporary and leased staff. Therefore, to ensure that our empirical analysis is still representative of the manufacturing sector, We construct propensity score weights based on a logit model of industry, firm size, and firm age to adjust the restricted sample to represent the LBD (Section A.3 provides details). The baseline sample for the analysis of this paper is a non-balanced panel of manufacturing plants between 2006 and 2017.

2.2 Domestic outsourcing in the cross-section

Table I describes the use of temporary and leased staff in the manufacturing sector by establishment age, size, and high-tech status. Column (1) displays the percentage of client plants: establishments reporting having spent any positive amount on outsourced employment, in a given year. Column (2) shows the amount spent on outsourced employment as a percentage of revenue for client plants.

The use of domestic outsourcing is common in the manufacturing sector. 70 percent of the plants reported having spent on outsourced employment at some point in time in the period between 2007 and 2017. Furthermore, 47 percent of the manufacturing plants every year are client plants.[9] The average client business spent 1.7 percent of its revenue on leasing labor through professional employment organizations and temporary help agencies.

Outsourcing arrangements are more prevalent in bigger and older establishments, but among clients, smaller and younger establishments use these arrangements more intensively. Panel A of Table I shows that the percentage of establishments with fewer than 10 payroll employees using staffing arrangements is one-third of that of establishments with more than 250 payroll employees. Conversely, among client establishments, small plants' temporary and leased staff share of revenue

[9]The difference indicates that using outsourced staff is not an absorbing state, though at any time a plant is more likely to spend on outsourced labor if it did so in the past. Manufacturing plants, in general, use outsourced staff intermittently.

Table I: Older, bigger, and high-tech establishments are more likely to spend on outsourced labor; conditional on doing so, however, younger, smaller, and low-tech establishments spend a larger share.[a]

	Share of clients	Share of revenue
Whole sample	**47.14**	**1.70**
Establishment age		
0-4	41.51	2.23
5-9	44.29	1.81
10-29	47.63	1.60
30+	51.97	1.45
Establishment size		
1-9	27.60	2.54
10-49	39.88	1.99
50-249	67.51	1.31
250+	81.19	1.01
High-tech status		
High-tech	63.62	1.46
Low-tech	46.11	1.72

[a]Entries reflect yearly averages within each establishment characteristic. Column 1 displays the percentage of establishments reporting having spent on temporary workers and leased employees. Column 2 reports the percentage of revenue spent on temporary and leased employees by the average client establishment in each category. High-tech status is defined through an establishment's 4-digit NAICS industry as in Hecker (2005). Source: Authors' calculations from ASM-CM-LBD data in 2006–2017.

more than double that of big establishments.

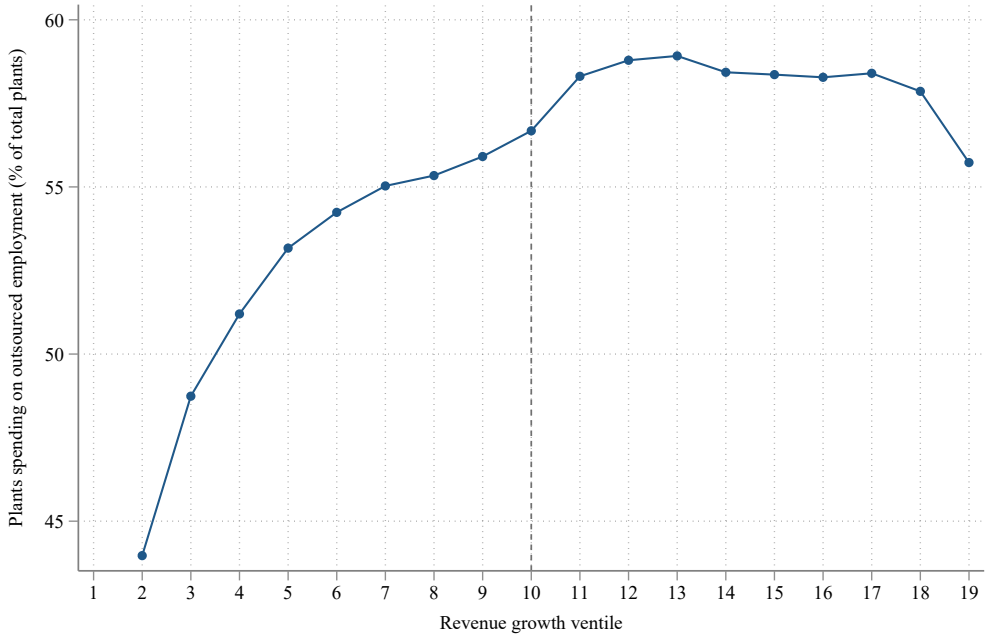
Haltiwanger, Hathaway and Miranda (2014) have found that the decline in payroll job reallocation between 2003 and 2011 was larger among high-tech manufacturing plants, so we investigate how outsourcing varies between high-tech and other manufacturing plants. High-tech is defined using four-digit NAICS industries as in Hecker (2005). Table I shows the results. Consistent with the omitted reallocations hypothesis, high-tech establishments are more likely to spend on staffing arrangements than other manufacturing plants. According to Table I, 63.6% of this type of manufacturing plant used staffing arrangements compared to 46.1% of low-tech establishments and 47.14% overall.

We next consider the cross-sectional relationship between the use of domestic outsourcing and average plant-level revenue growth. This exercise will provide insights on the strategic use of outsourced employment by plants. Our hypothesis is that the use of intermediaries allows for lower cost adjustments of labor needs and faster adjustment.

We restrict the sample to establishments observed for at least two consecutive years and compute

the symmetric revenue growth rate for each of them.[10] Then, we split the sample into twenty equally-sized groups and compute domestic outsourcing use statistics for each group. Figure 1 displays the average share of client establishments for each revenue growth ventile. Similarly, Figure 2 depicts average plant-level expenses on outsourced staff as a share of revenue for each revenue growth ventile.

Figure 1: The share of client plants increases with revenue growth. The increase flattens after the median.

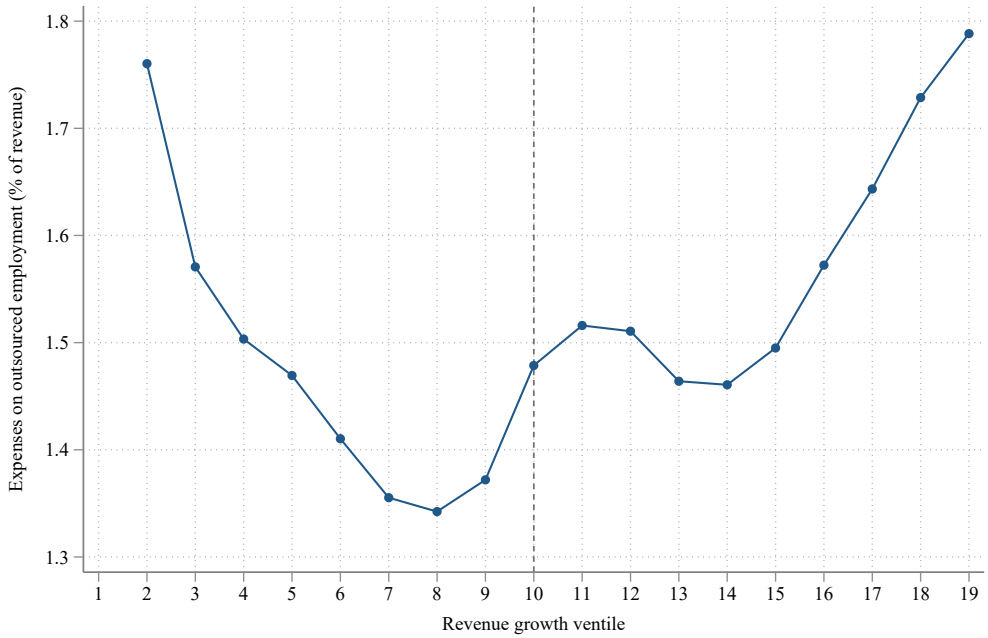


The figure displays the share of client establishment by revenue growth ventile. Each point is the three-point moving average. Source: Authors' calculation from ASM-CM-LBD data in 2006–2017.

Figure 1 shows an increasing, nonlinear relationship between revenue growth and the use of domestic outsourcing. The share of client plants declines sharply when revenue growth is shrinking. Above the median revenue growth, the participation margin rises until it reaches its maximum at 59.5%; then it flattens to finally decline for establishments at the top of the revenue growth distribution. Figure 1 indicates that the revenue growth and the decision of using outsourced workers are tightly linked at the business level. The "hook" shape of the relationship supports the hypothesis that plants use domestic outsourcing strategically and potentially as a margin of adjustment to variations in revenue growth. The pattern exhibited by the relationship between the intensive margin and revenue growth is consistent with this interpretation (Figure 2).

[10]DHS or arc-elasticity widely used in the literature. This measure is inclusive of establishments exiting or entering the market; however; we do not report them in figures 1 and 2.

Figure 2: There is a U-shaped relationship between the share of revenue spent on outsourced staff and revenue growth.



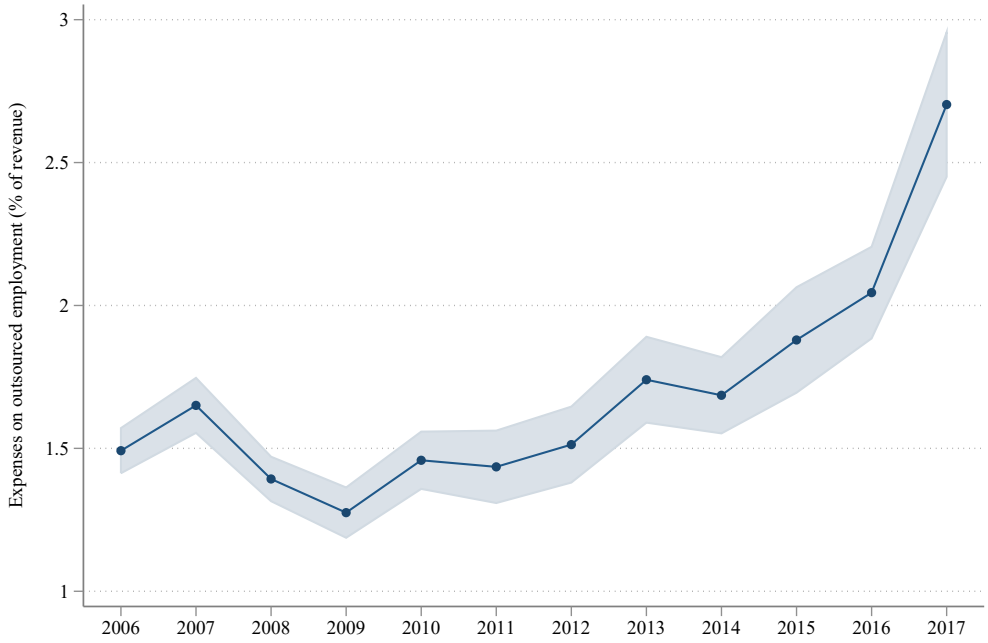
The figure displays the average share of revenue spent on temporary and leased staff by revenue growth ventile. Each point is the three-point moving average. Source: Authors' calculation from ASM-CM-LBD data in 2006–2017.

2.3 Domestic outsourcing over time

The use of domestic outsourcing is increasing over time. The share of revenue spent on this type of staffing arrangement almost doubled for the studied period, going from 0.65 percent to 1.21 percent. This growth is conditional on business age, industry, and payroll employment size, i.e., it is not accounted for changes in the composition of plants along these characteristics. The average client business exhibits the same pattern. Figure 3 depicts the temporary and leased staff share of revenue for the average client business over time. Every year, on average, plants that reported having spent a non-zero amount on temporary and leased staff increased the share of revenue spent on this service by approximately 7 percent, going from 1.49 percent in 2006 to 2.70 percent in 2017.

Despite the average positive yearly growth rate, between 2007 and 2009, the outsourced labor share of revenue dropped by 23 percent. During these years, the U.S. economy was undergoing the Great Recession; thus, the drop is expected but its magnitude is remarkable considering that revenue was also declining. It follows that client plants cut expenses on temporary and leased employment at a higher pace than they saw their revenue decline. Figure 4 suggests the same does not hold for payroll employment.

Figure 3: The average manufacturing plant increased the outsourced labor share by 85% between 2006 and 2017.

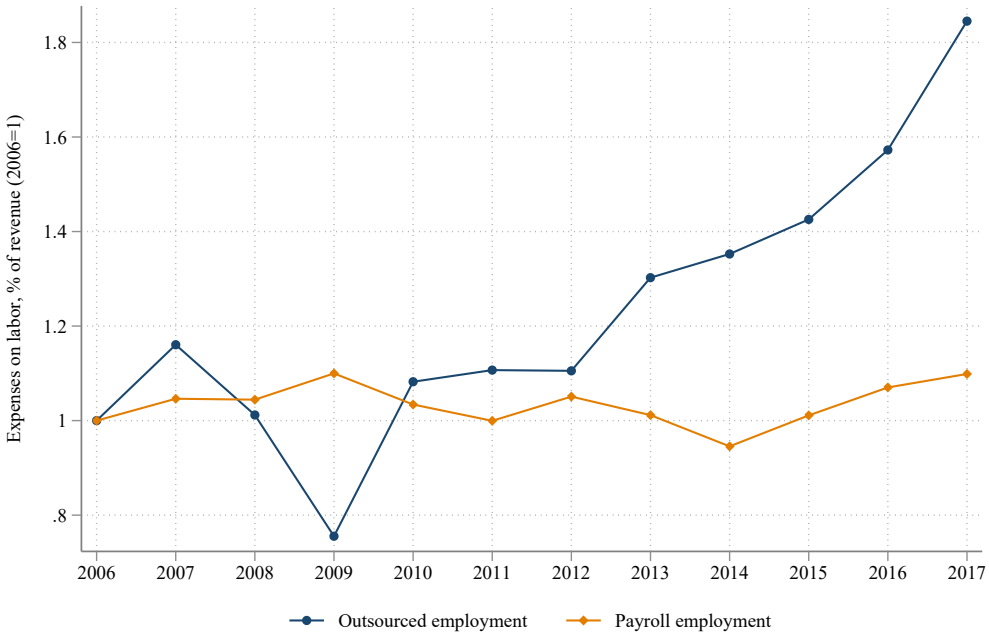


Point estimates and robust standard errors of plant-specific expenditures on outsourced staff as a share of revenue, controlling for employment size, age, and three-digit industry. Source: Authors' calculations from ASM-CM-LBD data in 2006–2017.

Figure 4 displays the labor share of outsourced and payroll employment for the average manufacturing business over time. It shows that, between 2007 and 2009, the average payroll employment share saw a small increase, presumably due to the decline in revenue, whereas the outsourced employment share dropped by 35%. Not surprisingly, the decline for the average manufacturing plants exceeds that of the average client business since the former confounds the fact that plants may decide not to use outsourced workers altogether in response to the adverse economic environment (the participation margin). This evidence is in line with the relationship between domestic outsourcing and revenue growth documented in the previous section. It also points to outsourced workers being an important margin of adjustment for plants. We investigate the role of outsourced workers as a margin of adjustment in section 3.

Figure 4 is also consistent with manufacturing plants increasingly sourcing labor from temporary help agencies and professional employment organizations even relative to direct hires. The figure additionally shows that outsourced labor share variance is significantly higher than that of payroll employment. The evidence suggests that the creation of jobs filled by outsourced workers has increased over time and supports the interpretation of outsourced workers as being an unobserved

Figure 4: Outsourced employment share of revenue is increasing, whereas payroll employment share of revenue remained roughly constant.



Point estimates of plant-specific expenditures on outsourced and payroll employment as a share of revenue, controlling for employment size, age, and three-digit industry. Point estimates normalized to the value in 2006. Source: Authors' calculations from ASM-CM-LBD data in 2006–2017.

margin of adjustment. Sections 3 and 4 confirm these statements.

3 Plant-level labor responsiveness

The omitted reallocations problem goes beyond measurement. The failure to account for outsourced workers translates into misconceptions about the labor market and how it works. We illustrate this point with the labor adjustment of plants to changes in revenue productivity growth (responsiveness). We show that by not considering outsourced workers in the labor responsiveness of plants we omit a margin of adjustment whose dynamics differ from that of the observed margin (payroll employment), limiting in turn our understanding of employers' strategic behavior. The increasing share of revenue spent on temporary and leased employment (see section 2) suggests that such an omitted margin of adjustment may be one of the underlying causes of the decline in payroll labor responsiveness documented by the literature.

3.1 Plant-level outsourced employment from reported expenses

Total expenditures on outsourced staff, denoted by exp^O, is the sum of the wage bill of outsourced workers, $o \times w^O$ plus fixed costs related to outsourcing F. That is, we have:

$$exp^O_{jst} = o_{jst} \cdot w^O_{jst} + F_{jst},$$

where plants, states and years are indexed by j, s and t, respectively.

Suppose that the fixed costs of outsourcing are proportional to the expenditure on outsourced employees, then we can write:

$$exp^O_{jst} = (1+\alpha) o_{jst} w^O_{jst}, \tag{1}$$

where α is the overhead per outsourced employee charged by the staffing agency associated with plant j.

To estimate plant-level outsourced employment o_{jst}, we begin by making two conservative assumptions. First, average earnings per payroll job w^p_{jst} are equal to that of outsourced jobs w^O_{jst}. Second, the agency's overhead per outsourced employee is equal to the inverse labor share β of the average staffing agency ℓ in state s.[11] Then, we estimate plant-level outsourced employment $\hat{o}$ as follows:

$$\hat{o}_{jst} = \underbrace{exp^O_{jst} \cdot \beta_{\ell st}}_{\text{outsourced wage bill}} \cdot \frac{1}{w^p_{jst}}. \tag{2}$$

Equation (2) underestimates plant-level outsourced employment as long as:

$$\beta_{\ell st} < \frac{1}{1+\alpha}, \tag{3}$$

Note that α is small, otherwise, the client plant would hire all employees directly instead of outsourcing. Moreover, besides competing with clients' direct hiring, staffing agencies also compete aggressively with each other on price. Consequently, $\frac{1}{1+\alpha} \approx 1$ and whenever outsourcing agencies have profits or expenses other than labor, $\beta_{\ell st} \ll 1$. Thus, condition (3) holds and our baseline estimate of plant-level outsourced employment is a lower bound.

Let us elaborate on the wage assumption. The average wage of outsourced workers is a fraction of that of payroll employees in the same occupation:

[11] We leverage location information on both staffing agencies and client plants to approximate an agency's labor share with the labor share of the average staffing agency in the state s in which the client plant is located β_{st} We do so because, although we observe the labor share of staffing agencies, the staffing agency-client plant link is not observed.

$$w_t^p = (1 + \gamma_t) w_t^O.$$

where γ_t denotes the time-variant wedge between payroll and outsourced workers. Dube and Kaplan (2010) show that for low-wage service occupations γ_t ranges from 4% to 24%. Using German data, Goldschmidt and Schmieder (2017) show there is an outsourcing wage penalty (ranging between 10% and 15%) even for jobs that are moved outside the boundary of the firm to contracting firms. Using data from the Occupation and Employment Wage Statistics (OEWS), We find that in our setting, γ_t varies from 10% to 15%. Equation (2) effectively assumes that γ_t is constant and equal to zero. Once again, our baseline results are a lower bound. We relax the wage assumption in section 4.

3.2 Outsourced employment and revenue productivity changes

We investigate outsourced workers as a margin of adjustment estimating a fixed-effects panel equation of outsourced employment growth on revenue productivity change ($log(a_t) - log(a_{t-1})$). Our empirical strategy is motivated by business dynamics models' result that plants adjust their employment in response to their own ever-changing productivity (Hopenhayn, 1992) and builds on the empirical design of Decker, Haltiwanger, Jarmin and Miranda (2020).[12] Specifically, we estimate the following equation:

$$\Delta_\tau y_j = \alpha_{it} + \beta_1 \Delta_t a_{jt} + \mathbf{X}'_{jt}\boldsymbol{\gamma} + \varepsilon_{j\tau} \tag{4}$$

where $\tau \in \{(t-1, t), (t, t+1), (t+1, t+2)\}$. Furthermore, α_{it} are six-digit industry-year fixed effects, j denotes establishments and t denotes year. The outcome $\Delta_\tau y$ is the symmetric (or DHS) growth rate of outsourced employment between any pair of years in τ.[13] The variable of interest is $\Delta_t a_{jt}$, the change in log revenue productivity between t and $t-1$. We estimate equation (4) for plants that stayed in the sample for at least five years.

Outsourced employment DHS growth rates, $\Delta_\tau y$, are inclusive of plants that stopped using this type of workers at any time during the studied period. This feature is important in our empirical design because plants may respond to revenue productivity changes by adjusting the intensity in which they use outsourced workers or they may stop using them altogether. DHS growth rates capture both responses computing outsourced employment annual changes relative to average outsourced employment for the two periods involved. However, this measure is not defined whenever such average is zero, i.e., non-client plants that decided to stay as such for the subsequent period.

[12] These authors study plant-level labor responsiveness as a cause for the decline in job reallocation.

[13] This growth rate concept is commonly used in the literature on firm dynamics. DHS refers to Davis, Haltiwanger and Schuh (1996).

We define the outcome as zero in these cases. In particular, for $\tau = (t-1, t)$

$$\Delta_\tau y_j = \begin{cases} \frac{\hat{o}_{jt} - \hat{o}_{jt-1}}{\frac{1}{2}(\hat{o}_{jt} + \hat{o}_{jt-1})} & \text{if } \hat{o}_{jt} + \hat{o}_{jt-1} > 0, \\ 0 & \text{otherwise} \end{cases}$$

where $\hat{o}_{jt}$ is the plant-level outsourced employment estimated using equation (2). Setting to zero the growth rate of plants that "dropped out" from using temporary and leased staff but did not exit the market allows me to capture, for example, situations in which the plant "hired" outsourced workers to fulfill a big order with a short deadline (less than one year) and then "laid them off": a common situation for the use of outsourced workers in manufacturing. The outcome would also capture the opposite case: plants that "opted in" from using the studied margin after the productivity change. It also permits me to follow the same set of establishments from $t-1$ to $t+2$, the periods involved in the estimation of equation (4).

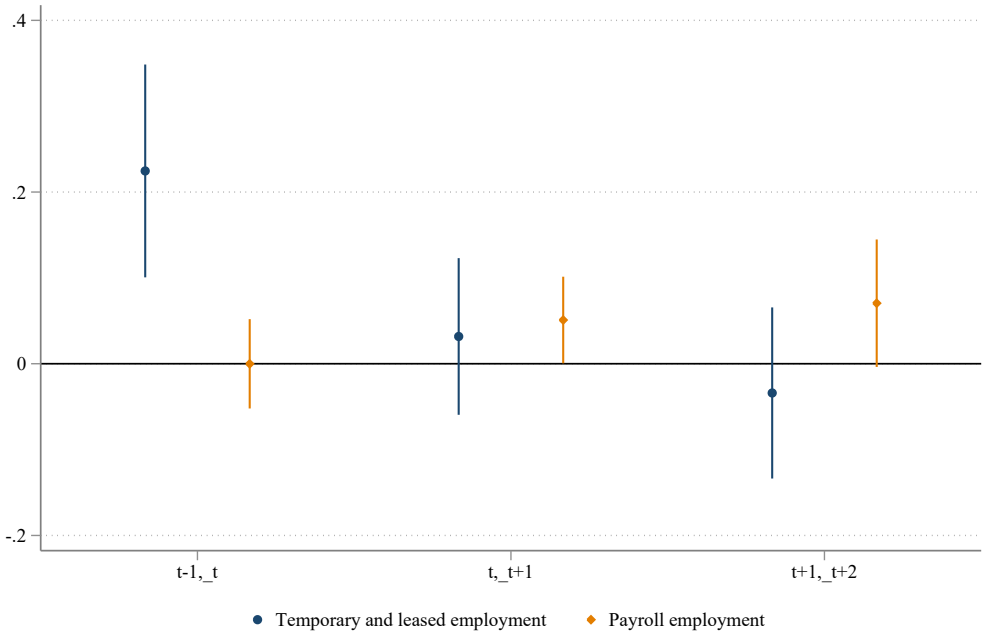
The parameter of interest, β_1, estimates outsourced labor responsiveness shocks. The average plant-level response of outsourced staff growth to *deviations* from industry-year average revenue productivity growth. The difference specification already nets out the estimated responsiveness of time-invariant factors at the establishment level; therefore, the inclusion of industry-year fixed effects makes β_1 the responsiveness to changes on plant-level deviations from the average revenue productivity growth in their detailed industry in a given year. We control for factors, common to all plants, inducing a linear trend in revenue productivity growth by including change in log productivity interacted with a linear trend in our set of controls (denoted by $\mathbf{X}_{jt}$).

In addition to the interaction between log revenue productivity change and a linear trend, our set of controls includes a variety of covariates informed by the results in sections 2 and 4. These include initial establishment log employment size, initial firm log employment size, firm age, and establishment age; in line with heterogeneity in the use of outsourced workers documented in section 2. The control set also includes a third-degree polynomial of log revenue productivity change, recognizing that the relationship between outsourced staff growth and revenue productivity growth is not linear as shown by Figure A.6. Other covariates in the control set are state fixed effects, ASM rotating sample fixed effects, and cyclical indicators: change in state unemployment rates, and change in state unemployment rates interacted with log revenue productivity change. We include cyclical indicators to avoid $\hat{\beta}_1$ being driven by the pro-cyclical feature of temporary and leased employment since the Great Recession is in the period of analysis.

We also estimate payroll labor responsiveness to deviations from average revenue productivity growth. Specifically, we estimate equation (4) with payroll employment DHS growth rate as the

outcome.[14] Figure 5 displays the estimated responsiveness of outsourced employment (blue) and payroll employment (red) to revenue productivity changes ($\hat{\beta}_1$), as well as 95% confidence intervals. Standard errors are clustered at the establishment level. The first column of Table II displays the results as well.

Figure 5: Plant-level outsourced employment growth is twice as responsive as payroll employment growth to revenue productivity shocks and adjusts more quickly.



The figure shows point estimates and 95-percent confidence intervals of plant-level payroll, and outsourced employment growth rate on revenue productivity change (see equation (4)). Source: Authors' calculations based on ASM-CM-LBD data and revenue-enhanced LBD from 2006–2017.

Plants use outsourced employment as a margin of adjustment. The response of outsourced employment DHS growth rate is sizable, statistically significant, and "immediate". For the average manufacturing business in the sample, a 1 percent deviation in its revenue productivity growth from the industry-year average is associated with a 0.22 percent increase in the outsourced employment growth rate in the same year of the productivity change. The comparable payroll employment responsiveness is half of that of outsourced employment.

Plants accelerate the use of outsourced workers only in the period of the shock (t), whereas their response through payroll employment occurs in the following period ($t + 1$). The correlation between the outsourced employment growth rate for the two periods following the productivity

[14] In contrast with outsourced employment, the average payroll employment between two periods (denominator of the DHS growth rate) is only zero if the business exited the market in the preceding year. This case is ruled out by the five-year sample restriction mentioned above.

change and the productivity change is small and not statistically different from zero. Conversely, the average business increases its payroll employment growth rate by 0.05 percent in $t+1$, when presumably it has more information on the persistence of the productivity change. Like outsourced employment, the payroll employment growth rate in $t+2$ is positive, but not statistically different from zero. Table II shows that the qualitative result holds for a different definition of the shock: deviations from own average productivity growth.

Table II: Plant-level outsourced employment growth is twice as responsive as payroll employment growth to revenue productivity shocks and adjusts more quickly.[b]

Dependent variable	Productivity shock $\Delta_\tau a_{jt}$		Mean
TYPE OF EMPLOYMENT GROWTH			
Payroll			
$t-1, t$	0.0001	−0.0521	2.66
	(0.0266)	(0.0111)	
$t, t+1$	0.05095	0.0286	2.57
	(0.0257)	(0.0145)	
$t+1, t+2$	0.07052	0.0072	3.79
	(0.0379)	(0.0173)	
Outsourced			
$t-1, t$	0.2246	0.2557	6.04
	(0.0633)	(0.0563)	
$t, t+1$	0.0318	0.0211	4.66
	(0.0466)	(0.0474)	
$t+1, t+2$	−0.0341	−0.0229	5.09
	(0.0509)	(0.0528)	
Fixed effects			
Industry-year	Yes		
Establishment		Yes	
OBSERVATIONS	102,000	102,000	

[b]The table contains plant-level labor responsiveness estimates to revenue productivity shocks. Standard errors are clustered at the establishment level and denoted in parentheses. Column (1) displays results for the baseline specification: shocks are defined as plants' revenue productivity growth deviations from 6-digit NAICS industry-year averages. Column (2) displays results when the shock is defined as plants' productivity growth deviations from their own productivity growth average. Column (3) displays the mean of the respective dependent variable. Source: Authors' calculations based on ASM-CM-LBD data.

Table III shows that the responsiveness result is robust to different specifications and not restricting the sample to five-period continuers.

The immediate response of outsourced staff reflects the flexible nature of this type of employment. Client plants can renegotiate staffing agreements within weeks. In fact, it is standard for staffing agencies to bill depending on the workers provided, allowing the client business to adjust tempo-

Table III: Outsourced employment growth is more responsive than payroll employment growth.[c]

Dependent variable	Mean	Level a_{jt}		Change $\Delta_\tau a_{jt}$	
TYPE OF EMPLOYMENT GROWTH					
Payroll $t, t+1$	-14.22	0.1782	0.1742	0.1061	0.1164
		(0.0110)	(0.0203)	(0.0395)	(0.0410)
Outsourced $t-1, t$	2.87			0.1374	0.1703
				(0.0467)	(0.0526)
Period					
1981–2013		Yes	Yes		
2006–2017				Yes	Yes
Controls					
Third-degree polynomial					Yes
Total initial employment					Yes
Industry-year fixed effects		Yes	Yes	Yes	Yes
OBSERVATIONS		1,630,000	1,181,000	150,000	150,000

[c]The table contains estimates for plant-level labor responsiveness estimates to revenue productivity shocks for the periods 1981–2013 and 2006-2017 and different regression specifications. Standard errors are clustered at the establishment level in parentheses. Column 1 contains average values for the dependent variable. Columns (2) and (3) display results for 1981–2013; the period of analysis in Decker, Haltiwanger, Jarmin and Miranda (2020). Columns (4) and (4) display results for 2006–2017; the period for which outsourced employment estimates are available. Column (2) reports payroll employment growth responsiveness to productivity levels following Decker, Haltiwanger, Jarmin and Miranda's 2020 specification. Column (3) reports payroll employment growth responsiveness to productivity changes following a specification otherwise identical to that of column (2). Column (4) reports additional specifications for outsourced employment growth responsiveness. Column (5) adds a third-degree polynomial in productivity and initial outsourced employment to the control set. All coefficients are statistically significant with at the 1 percent level. Source: Authors' calculations based on ASM-CM-LBD data.

rary and leased workers at a moment's notice, so that renegotiating the arrangement may not be even needed. We study yearly changes due to the frequency of the data available. The lagged plant-level response of outsourced staff relative to payroll staff is also in line with the pattern exhibited by the share of employment of outsourced workers at the aggregate level. This series drops sharply in the year preceding a recession as shown in Figure A.1. Therefore, this result provides micro-level evidence supporting the use of employment in the temporary help sector as a predictor of aggregate economic conditions. It also suggests that plants use outsourced and payroll employment as substitutes.

4 Secular decline in aggregate job reallocations

The previous section showed that plants use outsourced employment as a first-line margin of adjustment to shocks, and the increasing outsourced labor share documented in section 2 suggests this has been occurring at an increasing rate; thus, supporting the hypothesis that the increasing

use of domestic outsourcing is one of the underlying causes of manufacturing payroll employment becoming increasingly stable —payroll declining dynamism. In particular, the mismeasurement of labor market flows, as the dynamism is increasingly concentrated in the unmeasured mobility of outsourced workers. In this section, we show that, in fact, manufacturing outsourced jobs reallocate at a higher pace than payroll jobs. For job reallocations in the U.S. manufacturing sector between 2006 and 2017, we will show that the omitted reallocations problem is sizeable, exhibits considerable variation over time and across average plant-level revenue growth, and is tightly linked to the cycle.

We begin by defining job reallocations. Job reallocations capture the reshuffling of job opportunities across workplaces. Formally, they are the sum of plant-level employment gains and losses that occur between two years. Therefore, plant-level outsourced employment is the main ingredient to quantify how the creation and destruction of jobs filled by outsourced workers bias the measurement of the aggregate job reallocation. We calculate gross outsourced job reallocations OJR_t using plant-level outsourced employment $\hat{o}_{jt}$ estimated in section 3.1 as per equation (2):

$$OJR_t = \sum_{j \in \mathcal{J}_t} |\hat{o}_{jt} - \hat{o}_{jt-1}| \tag{5}$$
$$OJR_t = OJC_t + OJD_t$$

OJR_t captures the reallocation of outsourced jobs across manufacturing plants and it also equals the sum of the total number of outsourced jobs created OJC_t and destroyed OJD_t in a given year. Analogously, payroll job reallocations are the sum of payroll jobs created and destroyed in a given year.

Table IV summarizes the results. Panel A presents summary statistics for yearly outsourced jobs created, outsourced jobs destroyed, and outsourced gross job reallocations as a percentage of the corresponding payroll job flow between 2007 and 2017. They paint a clear picture: aggregate job flows are underestimated and the extent of mismeasurement varies significantly over time.

On average, every year, we omit the equivalent to 16 percent of the payroll jobs created and 13 percent of the payroll jobs destroyed. Both indicators display significant variation over time accounting for as much as one-fifth of the payroll job flow in a given year. Moreover, for the studied period, the share of outsourced jobs destroyed more than doubled, ranging from 8 to 20 outsourced jobs destroyed per every 100 payroll jobs destroyed. The reported variation is tightly linked to aggregate economic conditions, a point that we explore later in this section and that has non-trivial consequences on our understanding of labor market adjustment along the cycle.

Outsourced jobs reallocate at a higher pace than payroll jobs across plants. Panel B of Table IV

displays summary statistics for yearly outsourced and payroll job reallocation rates. That is, the corresponding job flow as a share of payroll or outsourced employment. For every job flow, the outsourced rate is at least two times higher than the corresponding payroll rate. The measurement of aggregate job flows not only omit the reallocations of a certain type of jobs, but these jobs also reallocate at a higher pace —a necessary condition for the omission of outsourced jobs reallocations to account for part of the documented decline in the payroll job reallocation rate. If the manufacturing sector were outsourcing longer-tenure jobs (relative to payroll jobs), the omitted reallocations problem would imply an overestimation of the payroll job reallocation rate.

Table IV: The omitted reallocations problem is sizeable and varies significantly over time.[d]

	JOB CREATION	JOB DESTRUCTION	JOB REALLOCATION
	(1)	(2)	(3)
	Panel A: Yearly outsourced job flow		
Average	16.20	13.66	**14.61**
SD	2.47	3.68	2.66
2017−2007	4.19	8.57	6.48
	Panel B: Yearly job flow rate		
Payroll			
Average	7.52	7.98	15.49
SD	1.29	2.86	2.07
2017−2007	12.31	−12.35	**−0.46**
Outsourced			
Average	30.38	25.99	56.36
SD	6.12	6.70	10.01
2017−2007	−1.42	6.21	**1.90**
Total			
Average	8.39	8.66	17.05
SD	1.38	2.88	2.11
2017−2007	14.81	−6.87	3.80
	Panel C: Job flow percentage change (2017−2007)		
Payroll	−2.14	−23.62	−13.27
Outsourced	22.99	32.50	27.12
Total	1.39	−17.76	−8.33
OBSERVATIONS	259,500	259,500	259,500

[d]Job creation is the sum of employment changes across expanding plants. Job destruction is the sum of employment changes across shrinking plants. Job reallocation is the sum of jobs created and jobs destroyed (see equation (5)). Outsourced job flows in panel *A* are expressed as a percentage of payroll job flows. Job flow rates in panel *B* are expressed as a share of employment. Source: Authors' calculations from ASM-CM-LBD data in 2006−2017.

For the studied period, the payroll job reallocation rate dropped by 0.46 percent. In contrast, the

outsourced job reallocation rate increased by 1.90 percent (Table IV, Panel B). Panel C of Table IV presents the percentage change in outsourced and payroll job reallocations.

We find that the documented increase in the outsourced job reallocation rate is driven by outsourced job reallocations increasing at a higher pace than outsourced employment (27.1 percent vs. 23.3 percent). Similarly, the drop in the pace at which payroll jobs reallocate across worksites is driven by payroll job reallocations decreasing at a higher pace than payroll manufacturing employment (13.3 percent vs. 11.9 percent). If the reallocation of outsourced jobs were considered in manufacturing job reallocations, the 13.27 percent decline would be 4.9 percentage points smaller. This is 37 percent of the drop.

The fact that outsourced employment increased by 23.3 percent while payroll employment dropped by 11.9 percent is yet another manifestation of the importance of outsourced workers for our understanding of labor markets. The employment trends documented in this section are not at odds with those in previous literature. In line with the representativeness of our analysis sample, the drop in manufacturing employment is comparable to that published by the Census Bureau for the manufacturing sector (11.5%). Moreover, reaffirming the validity of the methodology developed in this paper to estimate outsourced employment, the increase in outsourced employment in manufacturing between 2007 and 2017 is remarkably close to that found by Dey, Houseman and Polivka (2017) between 2007 and 2015 (23.1 percent). The authors follow a different methodology to estimate outsourced employment assigned to manufacturing combining employment-occupation information provided by staffing agencies (from the OEWS) with average occupation-industry of assignment distribution derived from worker-level data (CPS-CWS).

While confirming the increasing use of domestic outsourcing using client plant information, the results in this section also suggest a relevant role of domestic outsourcing *dynamics* for our understanding of labor markets' adjustment given the considerable variation exhibited by outsourced job flows.

The variation in the extent of aggregate job flows mismeasurement is tightly linked to aggregate economic conditions. Figure 6 displays omitted jobs created (green line) and destroyed (red line) as a share of the corresponding payroll job flow over time. The omitted job creation reached its maximum in 2010, implying that the total number of jobs created in manufacturing that year was 1.2 times the measured figure. In contrast, the omitted job destruction was at its minimum in the same year. This contrast means that, in the first year after the Great Recession, relative to payroll jobs, the manufacturing sector not only was creating jobs to be filled by outsourced workers at a higher pace but it was not destroying the existing ones as quickly. It follows that the omitted reallocations problem qualitatively affects our understanding of recoveries.

Figure 6: Aggregate job flows mismeasurement varies with the cycle. The share of omitted jobs created dropped entering the Great Recession and started increasing just before the recovery, reaching its maximum in 2010. The share of omitted jobs destroyed increased entering the recession and dropped just before the recovery, reaching its minimum in 2010.

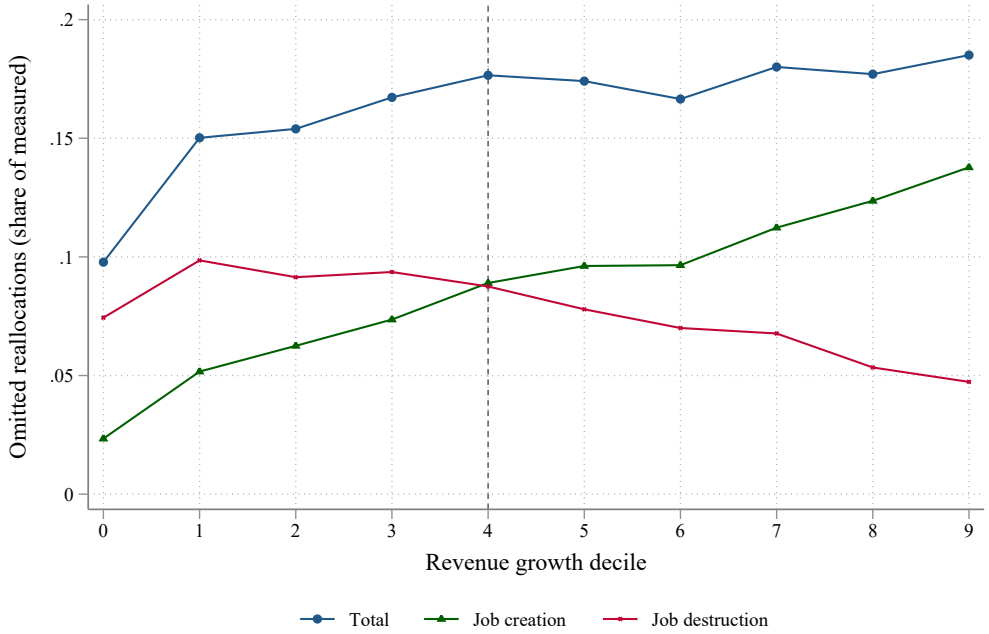


Job reallocations of outsourced employees as a share of the job reallocations of payroll employees over time. Source: Authors' calculations based on ASM-CM-LBD and RELBD from 2006-2017.

Since 2010, manufacturing plants destroyed payroll jobs at a slower pace than outsourced jobs, so that the omitted share of jobs destroyed more than doubled between 2010 and 2017 (from 8.4 percent to 20 percent). The same pattern does not hold for omitted job creation. Between 2011 and 2013, manufacturing plants created outsourced jobs at a faster pace than payroll jobs but the opposite is true between 2014 and 2017. This evidence is consistent with manufacturing plants handling the uncertainty in the aftermath of the Great Recession creating outsourced employment and then subsequently substituting it with increased payroll employment.

Figure 7 shows the average share of total outsourced reallocations by plant-level revenue growth decile. In general, the share of outsourced reallocations is increasing in revenue growth. However, the qualitative relationship between the share of outsourced jobs created and revenue growth contrasts starkly with that between the share of outsourced jobs destroyed and revenue growth: relative to the corresponding payroll job flow, outsourced job creation increases with revenue growth while outsourced job destruction decreases with revenue growth. On average, for plants experiencing negative (positive) revenue growth the share of omitted reallocations is mostly accounted for the

Figure 7: The destruction of jobs filled by outsourced workers accounts for most of the omitted reallocations in plants with negative revenue growth, while the creation of jobs filled by temporary and leased workers accounts for the omitted reallocations in plants with positive revenue growth.



The figure displays the average job reallocations of temporary and leased employees relative to measured job reallocations by revenue growth ventile. Each point is the three-point moving average. Source: Authors' calculations based on ASM-CM-LBD data and revenue-enhanced LBD from 2006–2017.

destruction (creation) of jobs filled by temporary and leased workers. This evidence further supports the interpretation that employers use outsourced workers strategically and shows that at the plant level, the sign of labor growth mismeasurement is not evident. At the aggregate level, gross job flows are undercounted; however, average plant-level employment growth might be under or overestimated depending on revenue growth.[15]

5 Conclusion

This paper shows that domestic outsourcing is becoming an increasingly more important margin of adjustment for U.S. manufacturing plants. Importantly, domestic outsourcing affects plant-level labor responses to revenue productivity shocks and biases the measurement of aggregate job

[15] If plants are outsourcing more expensive jobs over time, expenses on staffing services would exhibit the increasing trend documented in section 2 without translating into an increase in the number of temporary and leased workers employed in manufacturing. We address this concern in two ways. First, we show that the occupation distribution of temporary workers assigned to manufacturing in 2005 is comparable to that in 2017 (Table A.2). Second, the average earnings of temporary workers relative to that of payroll workers did not increase between 2007 and 2017, the period of our analysis.

reallocations. In particular, we assess the *omitted reallocations problem* on job reallocations in the U.S. manufacturing sector between 2006 and 2017. This issue arises when the hires and separations of outsourced workers, the creation and destruction of outsourced jobs, and the vacancies these workers fill, are not taken into account. Outsourced workers effectively work for client plants but are legally employed by a staffing agency; therefore, conventional data used to track labor markets' activity accounts for the labor market transitions of outsourced workers across staffing agencies, but it omits their reallocations across client plants altogether. This omission has two implications (i) a systematic undercount of the aggregate job and worker reallocations (and the vacancies they fill), and (ii) a misrepresentation of the composition of reallocations across sectors (Atencio De Leon, 2023).

The importance of the omitted reallocations problem hinges on the growth and prevalence of domestic outsourcing. Consequently, we describe the use of temporary and leased staff in the cross-section and over time. We document that the participation and intensity in the use of staffing services exhibit significant variation across size, age, industry, and revenue growth categories. Over time, the average manufacturing establishment increased the share of revenue spent on temporary and leased staff by 85 percent for the studied period. The growth of the outsourced labor share of revenue is eight times that of payroll and exhibits more variation.

At the sector level, the omitted reallocations problem undoubtedly leads to a systematic undercount of job reallocations, suggesting that part of the measured decline in job reallocations is a sign of a structural change in the way plants source labor instead of an actual decline in underlying dynamism. At the business level, however, the direction of the employment growth bias is not evident. We find that while the share of omitted jobs destroyed is declining along average plant-level revenue growth categories, the share of omitted jobs created is increasing. This evidence suggests that plant-level employment growth is overestimated whenever revenue is shrinking but underestimated when is growing. We investigate the omitted reallocations problem at the plant level (employment growth) studying the relationship between deviations from average revenue productivity growth and outsourced staff growth. We find that outsourcing workers is a margin of adjustment that reacts more quickly than the measured margin (adjustment of payroll employment). This provides plant-level micro-evidence for the use of temporary help employment as a leading business cycle indicator and calls for a more comprehensive measure to study the labor responsiveness of plants. Policies targeting firms commonly define eligibility and compliance conditions based on payroll jobs destroyed or created. However, this paper shows that plants may adjust labor through outsourcing. Typically, these outsourced workers have the same jobs as payroll workers and even work besides them at the client establishment.

Our micro-level findings on the plant-level characteristics of outsourcing and responsiveness have

significant aggregate consequences. The reallocation of jobs filled by temporary and leased workers across manufacturing plants accounts for 37 percent of the decline in payroll job reallocations. Hence, the omitted reallocations problem is sizeable, exhibits significant variation over time, and is tightly linked to aggregate economic conditions. The share of omitted jobs created peeked in 2010; the first year after the Great Recession. In contrast, the share of omitted jobs destroyed was at its minimum in the same year. Relative to payroll jobs, the manufacturing sector was not only creating jobs to be filled by outsourced staff at a faster pace, but it also was not destroying the existing ones. This evidence is indicative of the fact that we might have been underestimating economic recoveries and, thus, limits the effectiveness of recovery policies that are based on payroll employment.

According to evidence from the Contingent Worker Supplement of the Current Population Survey (CPS-CWS), manufacturing has been the biggest client industry since we have been collecting this information (i.e., 1995). Every year in which the CPS-CWS was conducted, around one-third of temporary help employees reported performing tasks in a manufacturing business (Appendix Table A.2). Interpreting domestic outsourcing as an innovation of the labor market, this evidence is consistent with manufacturing plants having learned how to incorporate this new "technology" in their production process faster than businesses in other industries. This could explain the lackluster amount of growth in manufacturing vis-à-vis other industries. Given the flexible nature of outsourced employment, we conclude that the magnitude of the omitted reallocations problem in the measurement of job reallocations may be even greater in sectors that started adopting this technology more recently.

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A Appendix

A.1 Additional Tables

Table A.1: The use of domestic outsourcing varies greatly across three-digit manufacturing industries.

	Pct. of establishments *(1)*	Pct. of revenue (clients) *(2)*
Food Manufacturing	47.33	1.81
Beverage and Tobacco	48.06	1.92
Textile Mills	51.03	1.60
Textile Product Mills	34.37	1.73
Apparel	26.98	1.81
Leather and Allied Product	33.03	1.27
Wood Product	37.17	2.11
Paper	70.78	1.14
Printing	44.14	1.93
Petroleum and Coal Products	25.36	1.27
Chemical	59.63	1.32
Plastics and Rubber	67.48	2.07
Nonmetallic Mineral	26.29	1.52
Primary Metal	59.24	1.05
Fabricated Metal	47.22	1.81
Machinery	52.69	1.46
Computer and Electronic	61.83	1.58
Electrical Equipment	62.44	1.44
Transportation Equipment	60.64	1.55
Furniture and Related	40.35	1.86
Miscellaneous	41.17	1.97
Total	47.14	1.70

Notes: Yearly averages by the given establishment characteristic. Column 1 displays the percentage of establishments reporting having spent on temporary workers and leased employees. Column 2 reports the percentage of revenue spent on temporary and leased employees by the average client establishment in each category. Industry groups correspond to the 3-digit NAICS classification. Source: Authors' calculations from ASM-CM-LBD data in 2006–2017.

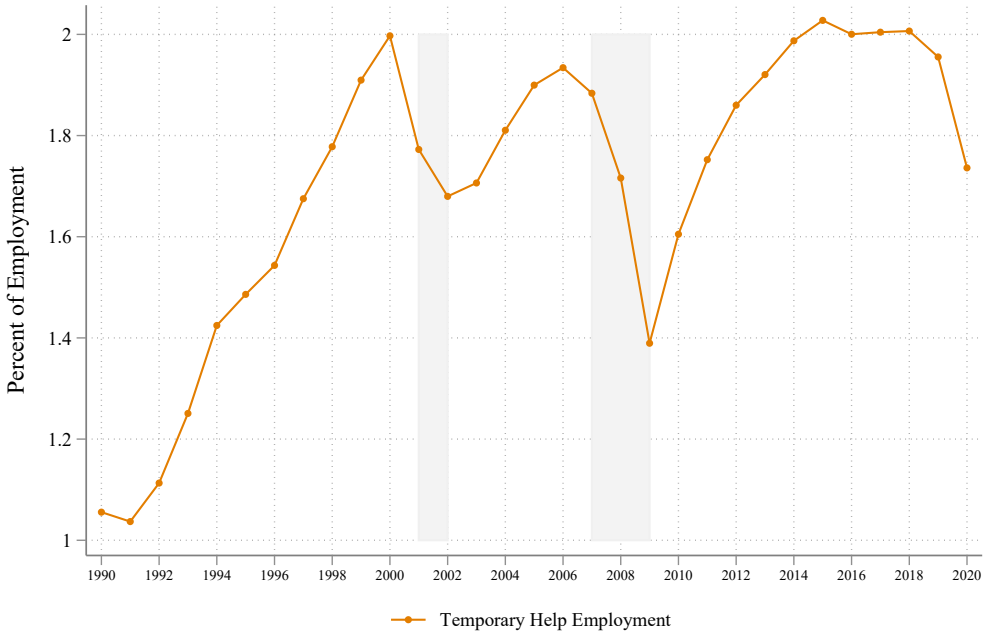
Table A.2: Industry of Assignment Distribution of Temporary Help Workers

	1995	1997	1999	2001	2005	2017
Agriculture, forestry, fishing	0.30	0.00	0.40	0.90	0.80	0.80
Mining	0.20	0.70	0.10	0.90	0.50	0.70
Construction	2.90	2.60	2.70	3.50	3.50	3.40
Manufacturing	34.10	32.10	31.20	22.70	29.50	34.90
Transportation, Communications	7.40	6.40	6.30	8.00	3.80	5.30
Wholesale trade	2.90	4.40	4.10	3.10	5.70	4.00
Retail trade	5.30	3.30	4.10	4.10	3.30	2.90
Finance, Insurance, and Real Estate	6.90	8.40	7.10	7.00	3.80	4.30
Business and repair services	22.60	25.90	25.60	30.30	29.20	23.20
Personal services	2.70	1.90	3.40	1.00	3.30	0.90
Entertainment and recreation services	0.70	0.90	0.50	1.90	0.00	0.60
Professional and related services	12.60	13.20	13.20	14.10	13.80	18.10
Public administration	1.30	0.00	1.20	2.40	2.90	1.00

Note: Calculations based on major industry of assignment (1990 codification) reported by those in the CWS who indicate being paid by a temporary help agency. CWS weights used. Source: Authors' calculations based on the CP-CWS.

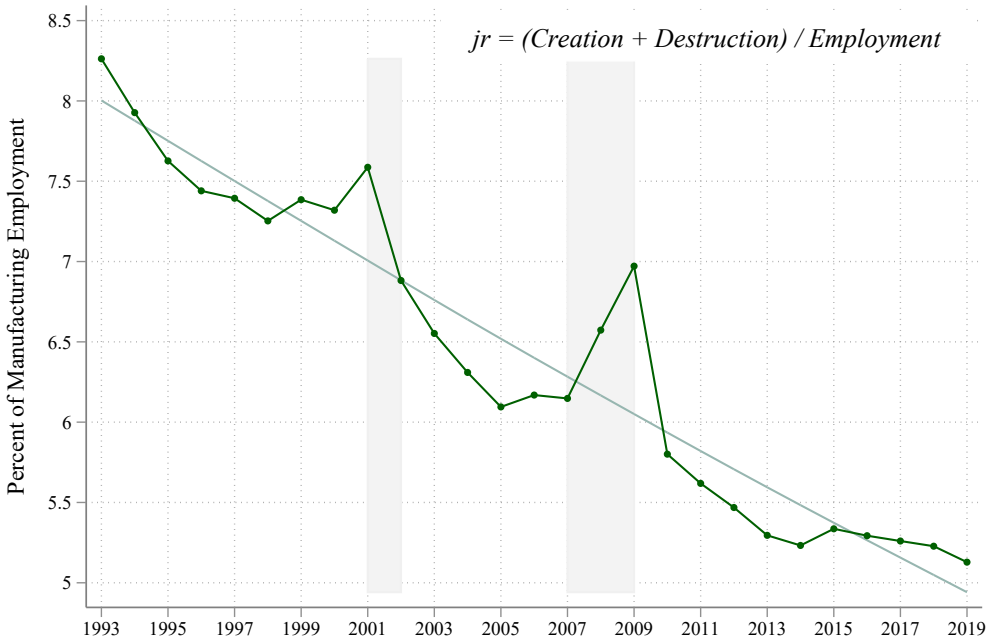
A.2 Figures

Figure A.1: The use of domestic outsourcing dramatically increased in the 1990s



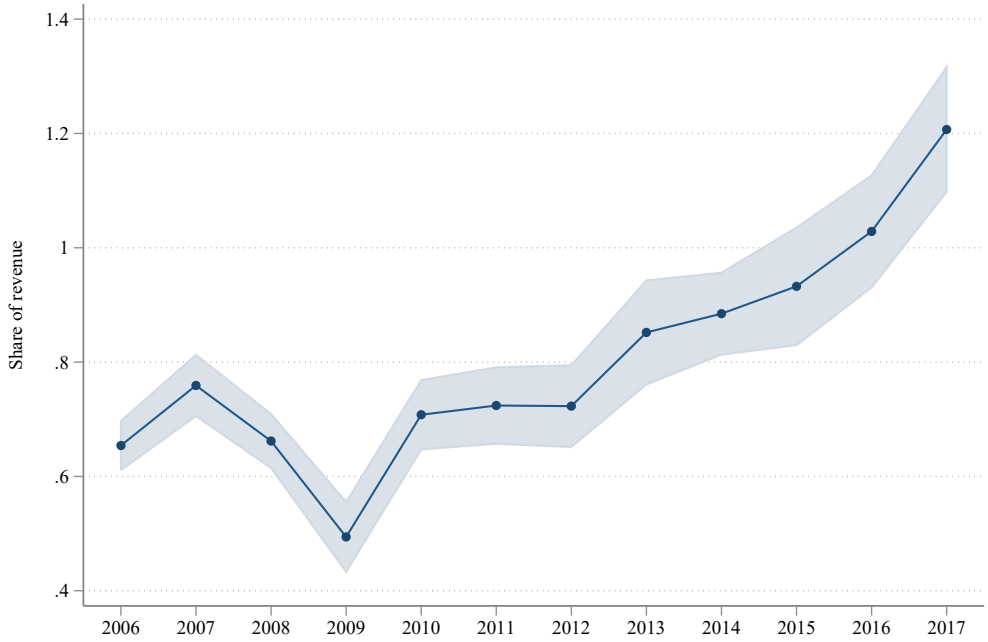
Note: The figure displays the average monthly employment in temporary help agencies relative to non-farm employment over time. Temporary help services is a six-digit NAICS industry comprised of establishments whose main activity is supplying workers to clients' plants. Source: Authors' calculations based on the Current Employment Statistics series, seasonally adjusted.

Figure A.2: The job reallocation rate in manufacturing has declined 40% since 1993.



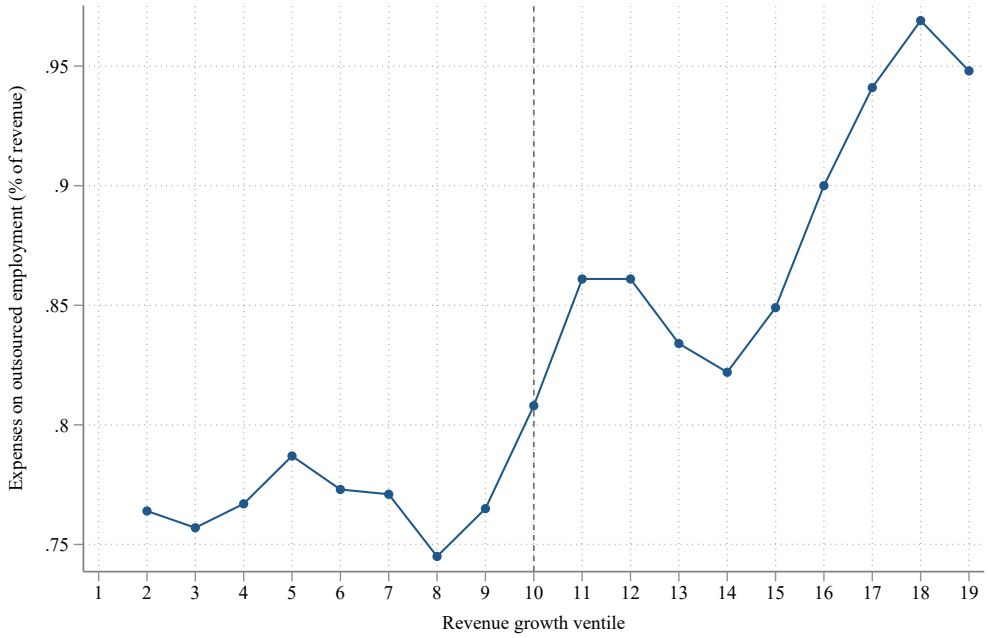
Note: The figure displays yearly averages of the quarterly job reallocation rate in the manufacturing sector and its HP trend. Source: Authors' calculations based on Quarterly Workforce Indicators, seasonally adjusted.

Figure A.3: The share of revenue spent on temporary and leased staff by the average establishment increased by 86% between 2006 and 2017.



Note: Table shows point estimates and robust standard errors of business-specific expenditures on temporary and leased staff as a share of revenue, controlling for employment size, age, and three-digit industry. Source: Authors' calculations from ASM-CM-LBD data in 2006–2017.

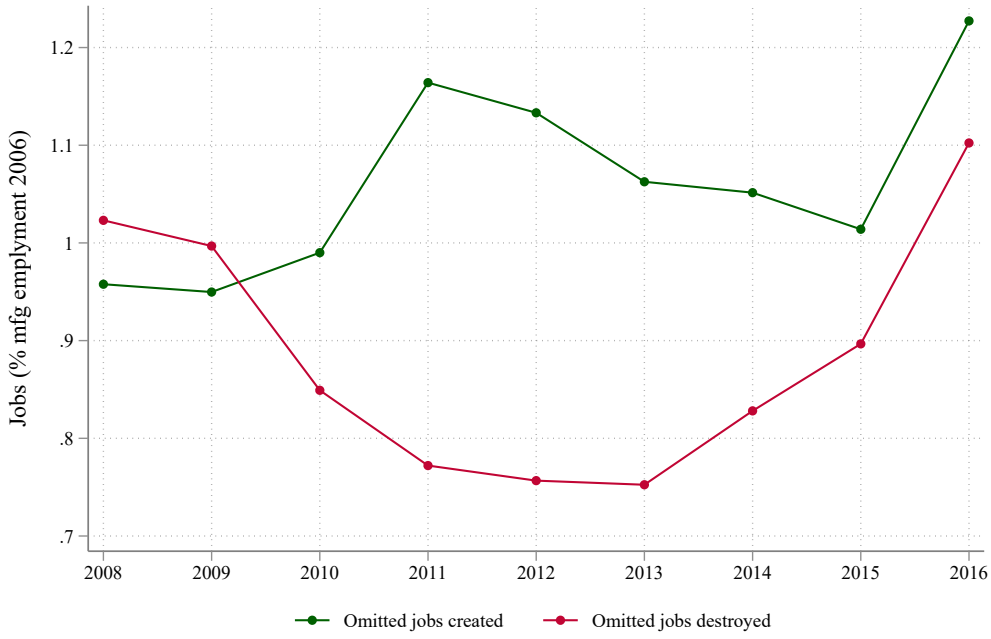
Figure A.4: Outsourced labor share along plant-level revenue growth for all establishments.



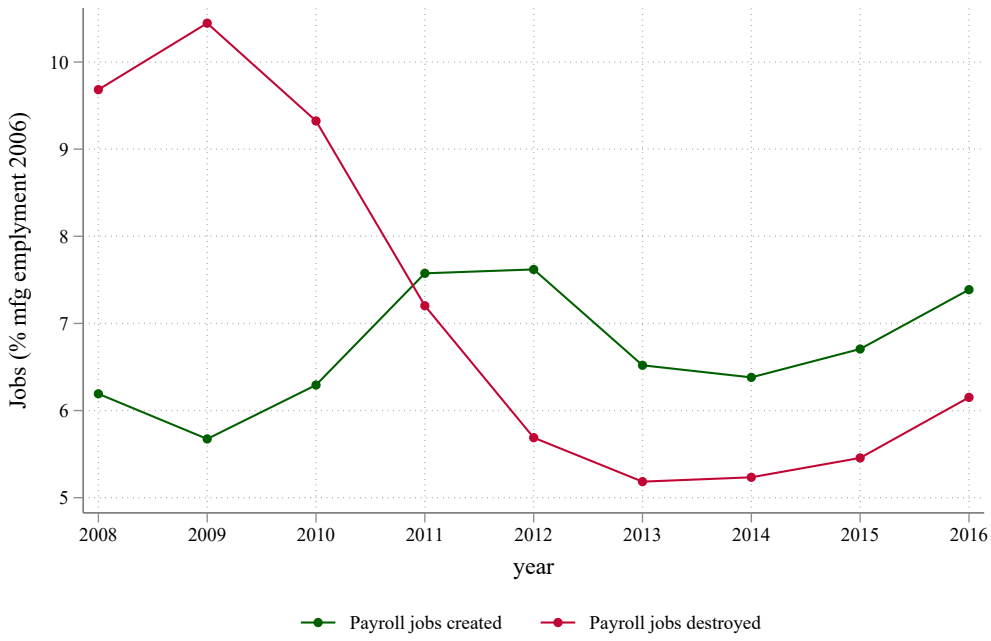
Note: The figure displays the average share of revenue spent on temporary and leased staff by revenue growth ventile. Each point is the three-point moving average. Source: Authors' calculation from ASM-SM-LBD data in 2006–2017.

Figure A.5: The manufacturing sector created more outsourced jobs than it destroyed

Panel A: Omitted job reallocations

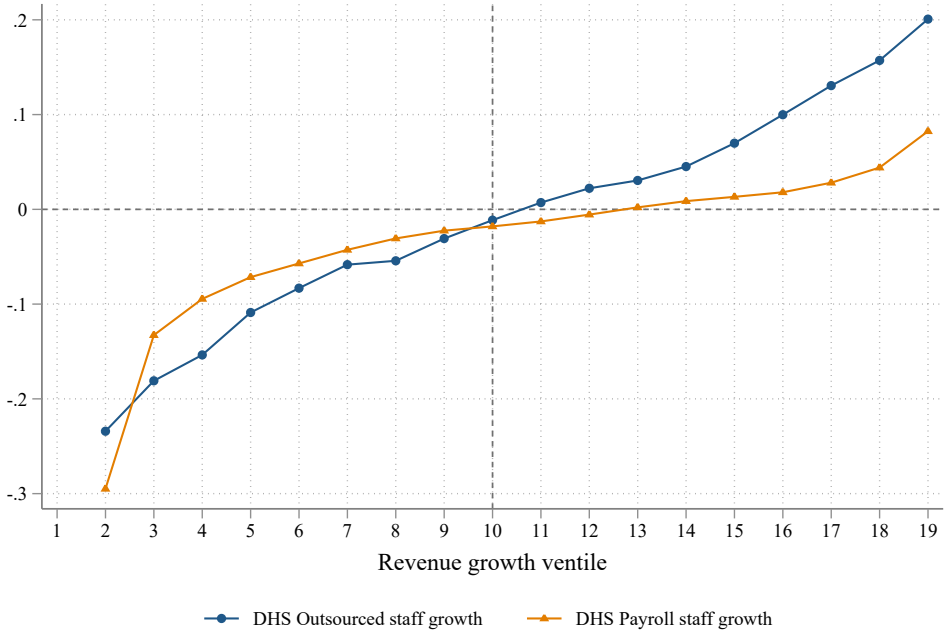


Panel B: Payroll job reallocations



Note: Each point is a three-year average. Omitted job reallocations computed following equation (5). Payroll job creation (destruction) is the sum of employment changes in expanding (shrinking) establishments. Source: Authors' calculation based on ASM-CM-LBD data and revenue-enhanced LBD from 2006–2017.

Figure A.6: The qualitative relationship between outsourced employment growth and revenue growth exhibits more variation than that of payroll employment.



Note: The figure displays the average DHS growth rate of temporary and leased employment and payroll employment by revenue growth ventile. Each point is the three-point moving average. Source: Authors' calculations based on ASM-CM-LBD data and revenue-enhanced LBD from 2006–2017.

A.3 Data appendix

Sample restrictions

We limit the analysis in Census years (years ending in 2 and 7) to those establishments that are part of the current rotating sample in the ASM. This is to ensure longitudinal consistency. Also, we drop observations that seem imputed. To do so, we identify plants whose payroll and cost of materials share (relative to revenues) deviate a lot from industry-level averages.

Exclusion criteria (based on other studies) (Dunne, 1998; Roberts and Supina, 1996)

1. Compute the ratio of total value of shipments and cost of materials to payroll for each establishment with a payroll greater than zero.
2. Drop establishments in which either of the ratios is zero or missing.
3. For each year in the sample, we drop establishments whose ratios equal the six-digit industry modal ratio.
4. For each year, we trim the industry-year TFP distribution by dropping establishments whose TFP deviate from the six-digit industry average by more than 2 in absolute value.
5. We delete establishments with zero or negative values in either of the TFP components: revenue, capital, total hours, materials and energy.
6. We winsorize capital at the 99.5 percentile.

Weights

We use an ASM-CM-LBD sample for our analysis. The ASM-CM provides the main variable of interest —expenses in outsourcing services, and the information to construct revenue productivity —main dependent variable. The LBD, on the other hand, has accurate establishment-level data on location, age, and firm characteristics.

We restrict the sample to establishments with information on outsourcing expenses. Therefore, to ensure that the analysis sample is representative of the manufacturing universe, we compute weights based on the probability of being sampled in the ASM and having reported outsourced expenses given that the establishment is in the LBD (through propensity scores). We run a logit regression in which the dependent variable is a dummy equal to one if the establishment is in both the ASM-CM and the LBD for that year and equal to zero if the establishment is only observed in the LBD. The independent variables are a multi-unit firm dummy, establishment size class dummies (measured by employment), payroll category dummies, and LBD detailed industry codes. Weights are then constructed through the inverse of the predicted logit probabilities.